

Large-Scale and Multi-Structured Databases

Project Design

Tonight's Call

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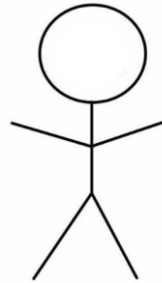
Alessandro Xavier Battisti

Application Highlights

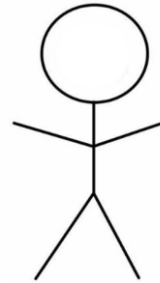
The aim of this application is to offer users a quick rundown of what events take place in their area (and beyond) and allow event organizers to publish their events

- *Users have access to a map to visualize every event based on their location*
- *Users have access to a browser to search for events based on topic, date or location*
- *Users can state their presence at events*
- *Users can review events and assign a numerical score to them*
- *Users can visualize events' details as well as organizers' profiles and ratings*
- *Organizers can create new events and publish them*
- *Organizers can work together in organizations and arrange events as a group*

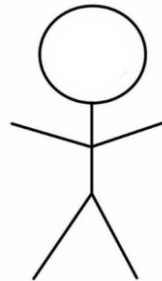
Actors and main mock-ups (i)



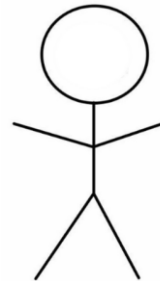
Unregistered
User



Registered
User



Single
Organizer



Organizing
Group

Actors and main mock-ups (ii)

Unregistered user

DISCOVER THE NEXT EVENTS IN THE WHOLE WORLD!

Welcome to the land of new events and opportunities! With us, you can discover the new events in your area to plan unforgettable days and nights with your friends and family or create and sponsor your own event... don't miss the chance to create some of your best memories!

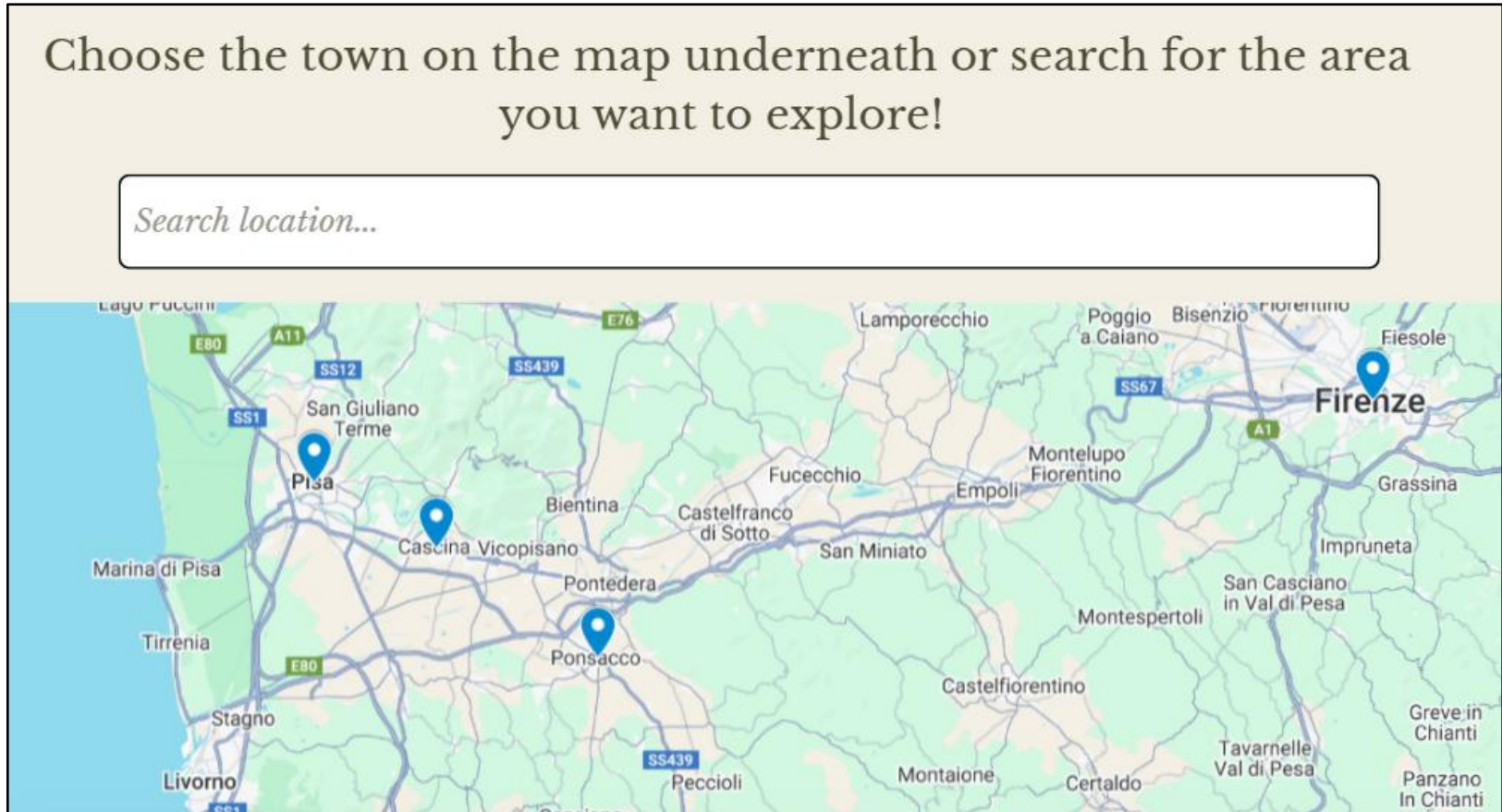
REGISTER

LOG IN

Main page where the user can create an account or log into their already existing one.

Actors and main mock-ups (iii)

Unregistered user



The user can explore the events by searching for a specific location in the search bar or by selecting a specific event on the map.

Actors and main mock-ups (iv)

Unregistered user

Select a theme and explore the next related events you could attend
or select a city and access its own events

Select theme

- food
- music
- history and culture
- animals
- nature
- art
- ...more

- Ancona
- Bari
- Bergamo
- Bologna
- Bolzano
- Brescia
- Cagliari
- Catania
- Ferrara
- Firenze
- Foggia
- Forlì
- Genova
- Giugliano in Campania
- Latina
- Livorno
- Messina
- Milano
- Modena
- Napoli
- Novara
- Palermo
- Parma
- Pavia
- Pescara
- Piacenza
- Prato
- Ravenna
- Salerno
- Sassari
- Siracusa
- Taranto
- Terni
- Torino
- Trento
- Trieste
- Venezia
- Verona

Page 1 of 10

The user can also explore the list of events in a specific city by clicking on it in the available list or search for events related to a selected theme.

Actors and main mock-ups (v)

Unregistered user

FESTIVAL DELLA FRAGOLA

Location
Terricciola (PI), Tuscany, Italy

Price
10-15 € per person

Date
3/5 - 11/5



The user can view the information about an event, like the location, the average price per person and the date(s) when it's holding.

Actors and main mock-ups (vi)

Unregistered user

The popular *Festival della fragola* in Terricciola, a little town near Pisa (Italy) is one of the most famous food festivals in Tuscany: you will have the chance to taste amazing farm-to-table strawberries, directly grown and collected on the fields surrounding the area. About the mean of transport, we recommend you travel by car or scooter, so that you will also be able to enjoy the view of the notorious Tuscanian hills where Terricciola was built.

If you need more information about this event, you can read some of the reviews left by other people who attended the festival by clicking the link underneath:

@emcarter_

Loved this festival! Never had tasted such amazing strawberries before! 5 ★

@jw_official

Definetely recommend! Delicious straweberries and friendly people! 4 ★

@parkersmiles

Expected better: price too high and place too crowded... 3 ★

[...more](#)

The user can also view a brief description of the event and reviews by people who have already attended it.

Actors and main mock-ups (vii)

Registered user

John Smith, 45

Cardiff, Wales (UK)

Food Music Nature

UPDATE PROFILE

Reviews 

Friend list 

Sagra della ciliegia, Lari (Italy) 2 

Lucca Summer Festival 2025, Lucca (Italy) 3 

Festival della fragola, Terricciola (Italy) 5 

@emcarter_

@jrw_official

@parkersmiles

Registered users' profile page, where they can update their own profile, view their friend list, read left reviews and write new ones.

 DIPARTIMENTO DI
INGEGNERIA
DELL'INFORMAZIONE

 UNIVERSITÀ DI PISA

 **CROSSLAB**
Innovation for industry 4.0

Actors and main mock-ups (viii)

Single Organizer

The mock-up shows a profile for Taylor Robinson, located in Pisa, Italy, with a 3.8 star rating. It includes buttons for 'UPDATE PROFILE' and 'JOIN ANOTHER GROUP'. Below the profile, there is a section for 'Events' with a green plus icon, listing three events: 'Concerto Gospel, Livorno (Italy)', 'Lucca Summer Festival 2025, Lucca (Italy)', and 'Festival della fragola, Terricciola (Italy)', followed by a link to see more events.

Taylor Robinson

Pisa, Italy

3.8 ★

[UPDATE PROFILE](#)

[JOIN ANOTHER GROUP](#)

Events +

Concerto Gospel, Livorno (Italy)

Lucca Summer Festival 2025, Lucca (Italy)

Festival della fragola, Terricciola (Italy)

[...more](#)

Organizers' profile page, where they can update their profile, join an organization, see the events they created or create a new one.

Actors and main mock-ups (ix)

Single Organizer / Organizing Group

FESTIVAL DELLA FRAGOLA	
Location	
Terricciola (PI), Tuscany, Italy	
Price	
10-15 € per person	
Date	
3/5 - 11/5	UPDATE DATA

The organizer can view and update the data related to an event they've created.

Actors and main mock-ups (x)

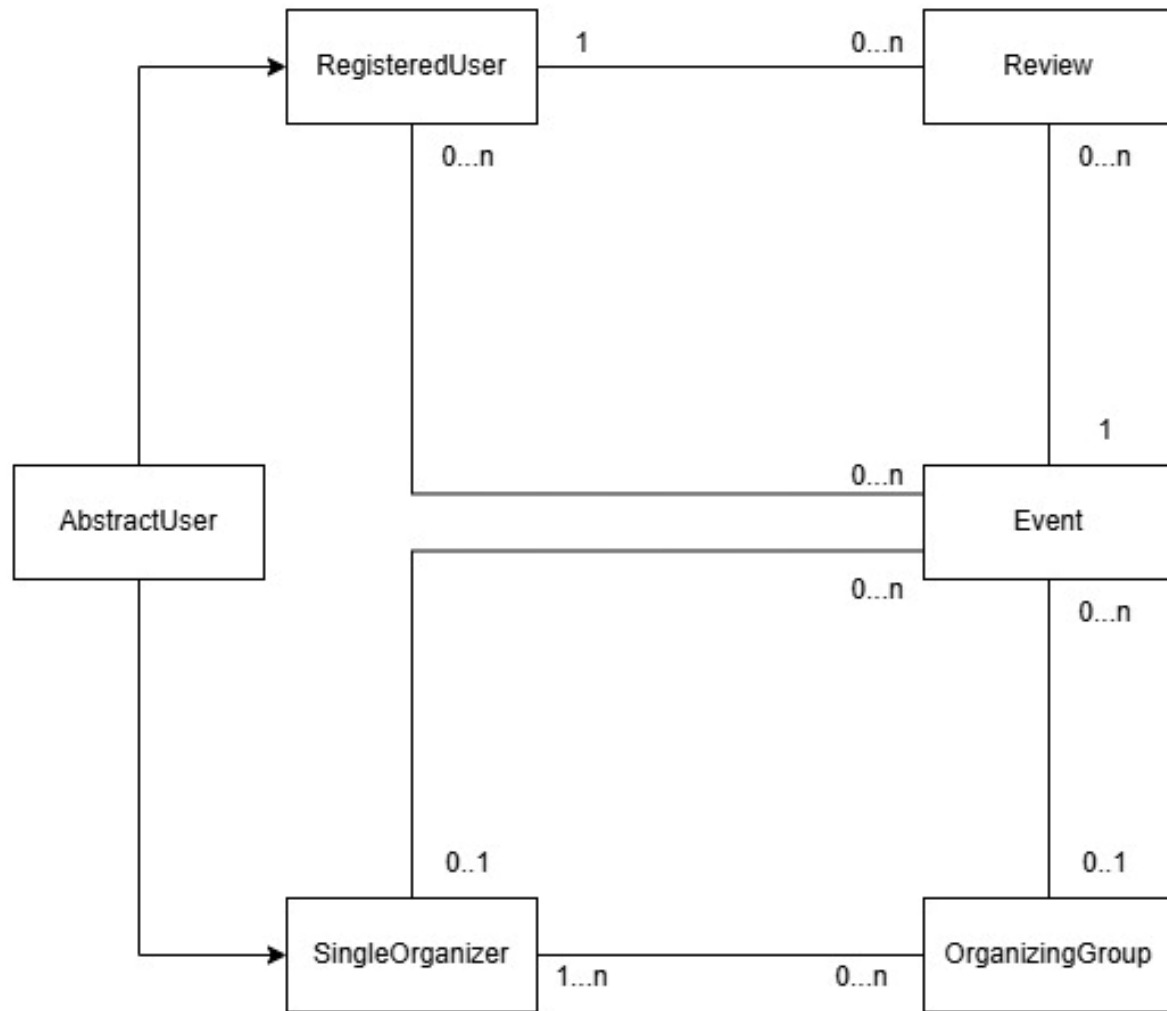
Single Organizer / Organizing Group

Statistics

- Average age: 53 years old
- Expected proceeds: 15 thousand (€)
- Total number of people: 5000
- People's geographical origin: Italy (80%), Germany (8%), France (12%)
- Average event's rating: 4.3 ★

The organizer can also see the statistics related to an event they have created.

Preliminary UML Class Diagram



Relevant Queries and Entities handled by Document DB

Functional Requirements

- The system must allow an unregistered user *to browse all the events based on location and/or theme and/or date*
- The system must allow registered users *to review events and assign them a score*
- The system must allow registered users *to visualize past event reviews and average scores of events and organizers*
- The system must be able to *show an event organizer various statistics for each event, such as:*
 - *Average age and provenance* of registered users who stated their presence at the event
 - *Predicted income*
 - *Average rating* of the event
- The system must allow an event organizer *to visualize the average scoring of his events*

Entities

- Event
- User
- Review

Requirements and Entities handled by Graph DB

Functional Requirements

- Whenever *a registered user states his presence to an event*, the registered users *in his friend list must receive a notification about it*
- The system must recommend a *registered user other events in which he might take part, based on:*
 - Previous reviews (for events which duration spans multiple days)
 - Past events the user has already attended to (mind the organizer of such events)
 - Distance from their home
 - Their own interests
- The system must *suggest users some possible friends based on interests and attended events*
- Whenever *an event organizer modifies an event's details*, the registered users *who stated their presence will receive a notification containing the modified information*

Requirements and Entities handled by Graph DB

